1. How many triplet codons constitute the standard genetic code?

(A) 16
(B) 32
(C) 64
(D) 128

2. Which codon serves as the universal translation initiation codon?

(A) UAA
(B) AUG
(C) UAG
(D) UGA

3. Which three codons are universal stop (nonsense) codons?

(A) UAA, UAG, UGA
(B) AUG, GUG, UUG
(C) UAA, UAU, UAC
(D) AGG, AGA, UGA

4. The genetic code is said to be 'degenerate' because:

(A) One codon codes for multiple amino acids
(B) Multiple codons can code for the same amino acid
(C) It varies between species
(D) It contains non-functional codons

5. Which amino acid is encoded by the standard start codon AUG?

(A) Methionine
(B) Tryptophan
(C) Leucine
(D) Valine

6. In prokaryotes, the initiating amino acid attached to initiator tRNA is:

(A) N-formylmethionine (fMet)
(B) Methionine
(C) Formylglycine
(D) Acetylmethionine

7. What is the sedimentation coefficient of the prokaryotic ribosome?

(A) 60S
(B) 70S
(C) 80S
(D) 100S

8. The prokaryotic 70S ribosome dissociates into which two subunits?

(A) 40S and 30S
(B) 50S and 30S
(C) 60S and 40S
(D) 50S and 20S

9. What is the sedimentation coefficient of the eukaryotic cytosolic ribosome?

(A) 70S
(B) 80S
(C) 60S
(D) 90S

10. The eukaryotic 80S ribosome dissociates into which two subunits?

(A) 50S and 30S
(B) 60S and 40S
(C) 50S and 40S
(D) 70S and 10S

11. Which rRNA molecule acts as the peptidyl transferase ribozyme in prokaryotes?

(A) 16S rRNA
(B) 23S rRNA
(C) 5S rRNA
(D) 18S rRNA

12. Which rRNA molecule acts as the catalytic peptidyl transferase in eukaryotes?

(A) 28S rRNA
(B) 18S rRNA
(C) 5.8S rRNA
(D) 5S rRNA

13. The ribosome has three tRNA binding sites: A, P, and E. What does 'A' stand for?

(A) Aminoacyl
(B) Peptidyl
(C) Exit
(D) Activation

14. The ribosome binding site on prokaryotic mRNA is termed the:

(A) Kozak sequence
(B) Shine-Dalgarno sequence
(C) Pribnow box
(D) TATA box

15. The Shine-Dalgarno sequence pairs with which component of the 30S subunit?

(A) 3' end of 16S rRNA
(B) 5' end of 23S rRNA
(C) 5S rRNA
(D) Ribosomal protein S1

16. In eukaryotic mRNA, optimal translation initiation is facilitated by the:

(A) Shine-Dalgarno sequence
(B) Kozak consensus sequence
(C) Poly-A tail only
(D) Pribnow sequence

17. Aminoacyl-tRNA synthetases catalyze the attachment of an amino acid to tRNA at the:

(A) 3' CCA-OH terminus
(B) 5' phosphate end
(C) Anticodon loop
(D) D-loop

18. How many high-energy phosphate bonds are consumed during tRNA charging?

(A) One (ATP -> ADP)
(B) Two (ATP -> AMP + PPi)
(C) Three
(D) Zero

19. The Wobble Hypothesis was originally formulated by:

(A) James Watson
(B) Francis Crick
(C) Sydney Brenner
(D) Marshall Nirenberg

20. According to the wobble hypothesis, unconventional base pairing occurs at the:

(A) 1st base of codon (3' of anticodon)
(B) 3rd base of codon (5' of anticodon)
(C) 2nd base of codon
(D) All positions equally

21. Which elongation factor delivers aminoacyl-tRNA to the ribosomal A-site in bacteria?

(A) EF-Tu
(B) EF-Ts
(C) EF-G
(D) IF2

22. Which factor regenerates active GTP-bound EF-Tu from EF-Tu-GDP?

(A) EF-Ts
(B) EF-G
(C) IF1
(D) RF1

23. Which elongation factor catalyzes ribosomal translocation in prokaryotes?

(A) EF-Tu
(B) EF-G
(C) EF-Ts
(D) RF2

24. The eukaryotic counterpart of bacterial EF-Tu is:

(A) eEF1A
(B) eEF1B
(C) eEF2
(D) eIF4E

25. The eukaryotic counterpart of bacterial EF-G is:

(A) eEF2
(B) eEF1A
(C) eIF2
(D) eIF3

26. Which release factor in E. coli recognizes UAA and UAG codons?

(A) RF1
(B) RF2
(C) RF3
(D) eRF1

27. Which release factor in E. coli recognizes UAA and UGA codons?

(A) RF1
(B) RF2
(C) RF3
(D) eRF1

28. In eukaryotes, how many release factors recognize all three stop codons?

(A) One (eRF1)
(B) Two (eRF1 and eRF2)
(C) Three
(D) Four

29. Which antibiotic inhibits bacterial translation by binding the 30S subunit and causing misreading?

(A) Streptomycin
(B) Chloramphenicol
(C) Erythromycin
(D) Puromycin

30. Which antibiotic mimics tyrosyl-tRNA and causes premature chain termination?

(A) Puromycin
(B) Tetracycline
(C) Chloramphenicol
(D) Rifampicin

31. Chloramphenicol inhibits bacterial protein synthesis by specifically blocking:

(A) Peptidyl transferase on 50S subunit
(B) EF-G translocation
(C) Initiation complex assembly
(D) tRNA synthetase

32. Diphtheria toxin inhibits eukaryotic translation by ADP-ribosylating:

(A) eEF2
(B) eIF4E
(C) eEF1A
(D) eIF2

33. The operon model for prokaryotic gene regulation was discovered by:

(A) Jacob and Monod
(B) Watson and Crick
(C) Hershey and Chase
(D) Meselson and Stahl

34. The lac operon is an example of an:

(A) Inducible operon
(B) Repressible operon
(C) Constitutive operon
(D) Attenuated operon only

35. The natural inducer that binds the Lac repressor is:

(A) Allolactose
(B) Lactose
(C) Glucose
(D) Galactose

36. A synthetic, non-metabolizable inducer widely used in molecular biology is:

(A) IPTG
(B) X-gal
(C) cAMP
(D) ONPG

37. The lacZ gene encodes which enzyme?

(A) Beta-galactosidase
(B) Beta-galactoside permease
(C) Thiogalactoside transacetylase
(D) Lac repressor

38. The lacY gene encodes which protein?

(A) Beta-galactoside permease
(B) Beta-galactosidase
(C) Transacetylase
(D) CAP

39. The trp operon is an example of a:

(A) Repressible operon
(B) Inducible operon
(C) Constitutive operon
(D) Non-regulated operon

40. In the trp operon, tryptophan functions as a:

(A) Corepressor
(B) Inducer
(C) Apo-repressor
(D) Activator

41. Which base in the tRNA anticodon allows wobble pairing with U, C, and A?

(A) Inosine (I)
(B) Pseudouridine
(C) Dihydrouridine
(D) Queuosine

42. Aminoacyl-tRNA synthetases ensure fidelity through a 'double-sieve' editing mechanism. The second sieve:

(A) Hydrolyzes incorrectly charged aminoacyl-tRNAs
(B) Prevents large amino acids from entering
(C) Binds mRNA codons
(D) Phosphorylates tRNA

43. Class I and Class II aminoacyl-tRNA synthetases differ fundamentally in:

(A) Class I attaches to 2'-OH; Class II attaches to 3'-OH of terminal adenosine
(B) Class I uses GTP; Class II uses ATP
(C) Class I binds only mRNA
(D) Class I is ribosomal

44. Which initiation factor prevents premature reassociation of 30S and 50S subunits in bacteria?

(A) IF3
(B) IF1
(C) IF2
(D) EF-Tu

45. Which bacterial initiation factor is a GTPase that escorts fMet-tRNA to the partial P-site?

(A) IF2
(B) IF1
(C) IF3
(D) EF-G

46. Which eukaryotic initiation factor binds the 5' 7-methylguanosine cap of mRNA?

(A) eIF4E
(B) eIF4G
(C) eIF4A
(D) eIF2

47. Which eukaryotic factor acts as an ATP-dependent RNA helicase to unwind 5' secondary structure?

(A) eIF4A
(B) eIF4E
(C) eIF4G
(D) eIF3

48. Phosphorylation of the alpha subunit of eIF2 by kinases such as PERK or PKR leads to:

(A) Global translational inhibition
(B) Accelerated translation initiation
(C) Increased ribosome biogenesis
(D) Immediate mRNA degradation

49. 4E-BP1 represses translation initiation by competing with which factor for binding to eIF4E?

(A) eIF4G
(B) eIF4A
(C) eIF3
(D) eIF2

50. Phosphorylation of 4E-BP1 by which master signaling kinase promotes translation initiation?

(A) mTORC1
(B) PKA
(C) AMPK
(D) GSK3

51. Ribosomal translocation requires the energy of:

(A) GTP hydrolysis by EF-G (or eEF2)
(B) ATP hydrolysis by helicase
(C) Phosphocreatine
(D) Proton motive force

52. During peptide bond formation, the alpha-amino group of the A-site amino acid attacks the:

(A) Carbonyl carbon of the peptidyl-tRNA in the P-site
(B) Phosphate backbone of 23S rRNA
(C) CCA terminus of A-site tRNA
(D) Carboxyl group of free methionine

53. The ribosome displays 'hybrid states' (A/P and P/E) during translation. This transition occurs:

(A) After peptide bond synthesis but before EF-G-mediated translocation
(B) Before codon-anticodon pairing
(C) During ribosome recycling only
(D) During initiation

54. The Ribosome Recycling Factor (RRF) in bacteria acts in concert with which factor to split subunits?

(A) EF-G
(B) EF-Tu
(C) IF3
(D) RF3

55. In bacterial catabolite repression, when glucose levels are high, cellular cAMP levels are:

(A) Low, preventing CAP-cAMP binding to the promoter
(B) High, activating CAP
(C) Unchanged
(D) Oscillating rapidly

56. The binding of cAMP-CAP complex to the lac promoter causes:

(A) A 90-degree bend in DNA, recruiting RNA polymerase
(B) Repression of transcription
(C) Destruction of lac repressor
(D) Degradation of mRNA

57. Attenuation in the trp operon is governed by:

(A) Coupling of transcription and translation at the leader peptide
(B) Binding of Trp repressor to the operator
(C) Ubiquitination of RNA polymerase
(D) Alternative sigma factors

58. When cellular tryptophan is abundant, which hairpin structure forms in the trp leader RNA?

(A) 3:4 terminator hairpin
(B) 2:3 antiterminator hairpin
(C) 1:2 promoter hairpin
(D) 1:4 pseudoknot

59. When cellular tryptophan is scarce, the ribosome stalls at Trp codons, permitting formation of the:

(A) 2:3 antiterminator hairpin
(B) 3:4 terminator hairpin
(C) 1:3 loop
(D) 4:5 attenuation stem

60. Riboswitches are regulatory RNA elements typically located in the:

(A) 5' untranslated region (5'-UTR) of mRNA
(B) Coding sequence
(C) 3'-poly-A tail
(D) Introns

61. A riboswitch consists of two structural domains: an aptamer domain and an:

(A) Expression platform
(B) Anticodon loop
(C) Active catalytic center
(D) Enhancer element

62. Which core histone modification is universally associated with transcriptional activation?

(A) Histone H3 acetylation (e.g., H3K9ac, H3K27ac)
(B) Histone H3 deacetylation
(C) H3K9 trimethylation (H3K9me3)
(D) H3K27 trimethylation (H3K27me3)

63. Histone Acetyltransferases (HATs) neutralize the positive charge of which amino acid residue?

(A) Lysine
(B) Arginine
(C) Histidine
(D) Serine

64. Histone Deacetylases (HDACs) repress transcription by:

(A) Restoring the positive charge on lysines, promoting chromatin compaction
(B) Degrading histone tails
(C) Methylating DNA at CpG islands
(D) Phosphorylating RNA Pol II

65. Bromodomains present in chromatin remodeling complexes specifically recognize:

(A) Acetylated lysine residues
(B) Methylated lysine residues
(C) Phosphorylated serine residues
(D) Ubiquitinated histones

66. Chromodomains specifically recognize and bind to:

(A) Methylated lysine residues (e.g., H3K9me3, H3K27me3)
(B) Acetylated lysines
(C) Phosphorylated serines
(D) ADP-ribose polymers

67. DNA methylation in mammals predominantly occurs at cytosine bases in the context of:

(A) CpG dinucleotides
(B) GpC dinucleotides
(C) UpA pairs
(D) Poly-T tracts

68. Which enzyme catalyzes maintenance DNA methylation following replication?

(A) DNMT1
(B) DNMT3A
(C) DNMT3B
(D) TET1

69. Which enzymes catalyze de novo DNA methylation during embryogenesis?

(A) DNMT3A and DNMT3B
(B) DNMT1 and DNMT2
(C) TET2 and TET3
(D) HDAC1 and HDAC2

70. CpG island hypermethylation in promoter regions typically leads to:

(A) Stable transcriptional silencing
(B) Constitutive transcription activation
(C) Increased translation rate
(D) Enhanced splicing

71. Enhancer sequences can regulate target promoters across large genomic distances through:

(A) Chromatin looping mediated by cohesin and CTCF
(B) Linear sliding along naked DNA
(C) Passive diffusion of RNA polymerase
(D) Nuclear envelope degradation

72. The zinc finger motif coordinates a zinc ion typically using which amino acid combinations?

(A) Two Cysteines and two Histidines (C2H2)
(B) Four Arginines
(C) Two Lysines and two Aspartates
(D) Four Methionines

73. The basic Leucine Zipper (bZIP) motif mediates transcription factor dimerization via:

(A) Heptad repeats of hydrophobic leucine residues
(B) Zinc coordination
(C) Disulfide bridges
(D) Proline-rich beta-hairpins

74. The Helix-Turn-Helix motif interacts with DNA primarily through which structure?

(A) Recognition helix inserted into the major groove
(B) Beta-barrel in minor groove
(C) Disordered loop in phosphate backbone
(D) Leucine coiled-coil

75. The primary enzyme complex responsible for microRNA processing inside the nucleus is:

(A) Drosha - DGCR8 (Microprocessor)
(B) Dicer - TRBP
(C) Argonaute2 - RISC
(D) Exosome

76. Following nuclear export by Exportin-5, pre-miRNA hairpins are cleaved in the cytoplasm by:

(A) Dicer
(B) Drosha
(C) Argonaute 1
(D) XRN1

77. The core catalytic engine of the RNA-induced Silencing Complex (RISC) that possesses 'slicer' activity is:

(A) Argonaute 2 (AGO2)
(B) DGCR8
(C) TRBP
(D) PABP

78. In animal cells, imperfect base pairing of miRNA with the 3'-UTR of target mRNA primarily causes:

(A) Translational repression and deadenylation-dependent mRNA decay
(B) Endonucleolytic cleavage of mRNA
(C) Target DNA methylation
(D) Alternative splicing

79. Small interfering RNAs (siRNAs) typically trigger endonucleolytic cleavage because they exhibit:

(A) Near-perfect Watson-Crick complementarity to target mRNA
(B) Extensive G:U wobble mismatches
(C) A single-strand loop structure
(D) Poly-uridine tails

80. Long non-coding RNA Xist orchestrates X-chromosome inactivation by recruiting which complex?

(A) Polycomb Repressive Complex 2 (PRC2)
(B) SWI/SNF complex
(C) Basal transcription factor TFIID
(D) Histone acetyltransferase p300

81. The catalytic subunit of PRC2 that methylates histone H3 at lysine 27 (H3K27me3) is:

(A) EZH2
(B) SUZ12
(C) EED
(D) RbAp48

82. In prokaryotic translation, the tmRNA (SsrA RNA) system rescues ribosomes stalled on broken mRNAs by:

(A) Acting as both tRNA and mRNA to add a degradation tag
(B) Cleaving the 50S subunit
(C) Degrading all ribosomes
(D) Inhibiting IF2

83. The C-terminal peptide tag encoded by tmRNA targets defective proteins for destruction by:

(A) ClpXP and ClpAP proteases
(B) 20S proteasome
(C) Lysosomal hydrolases
(D) Caspase-3

84. Nonsense-Mediated mRNA Decay (NMD) in mammalian cells is triggered when a stop codon is located:

(A) More than 50-55 nucleotides upstream of an Exon Junction Complex (EJC)
(B) At the very 3' end of the transcript
(C) Within the first exon
(D) Inside the poly-A tail

85. The key phosphoprotein kinase that coordinates Upf1 activation during NMD is:

(A) SMG1
(B) ATM
(C) ATR
(D) mTOR

86. Ubiquitin is covalently linked to target protein substrates at which amino acid residue?

(A) Epsilon-amino group of Lysine
(B) Carboxyl group of Glutamate
(C) Hydroxyl group of Serine
(D) Guanidinium group of Arginine

87. The canonical enzymatic cascade of ubiquitination follows which sequential order?

(A) E1 (Activating) -> E2 (Conjugating) -> E3 (Ligase)
(B) E3 -> E2 -> E1
(C) E2 -> E1 -> E3
(D) E1 -> E3 -> E2

88. Polyubiquitin chains linked through which specific lysine residue target substrates to the 26S proteasome?

(A) Lysine 48 (K48)
(B) Lysine 63 (K63)
(C) Lysine 11 (K11)
(D) Lysine 29 (K29)

89. Polyubiquitin chains linked through Lysine 63 (K63) primarily function in:

(A) DNA repair signaling, endocytosis, and NF-kB activation
(B) Proteasomal degradation
(C) Ribosome biogenesis
(D) Nuclear export

90. The 26S proteasome is composed of which two structural complexes?

(A) 20S core catalytic barrel and 19S regulatory caps
(B) 10S core and 16S caps
(C) 30S subunit and 50S subunit
(D) 40S core and 60S cap

91. The catalytic active sites within the beta-rings of the 20S proteasome utilize which nucleophilic residue?

(A) N-terminal Threonine (Thr1)
(B) Serine
(C) Cysteine
(D) Aspartate

92. Internal Ribosome Entry Sites (IRES) allow translation initiation to occur:

(A) Independently of the 5' 7-methylguanosine cap
(B) Without any ribosomal subunits
(C) Without initiator tRNA
(D) Exclusively in the nucleus

93. During integrated stress response, phosphorylation of eIF2-alpha selectively UP-REGULATES translation of:

(A) ATF4 mRNA via upstream open reading frames (uORFs)
(B) Actin mRNA
(C) GAPDH mRNA
(D) All ribosomal proteins

94. P-bodies (Processing bodies) in the cytoplasm are specialized ribonucleoprotein granules enriched in:

(A) mRNA decapping enzymes (Dcp1/Dcp2) and 5'->3' exonuclease Xrn1
(B) Active polysomes and translation factors
(C) Spliceosomal snRNPs
(D) DNA polymerases

95. Stress granules differ from P-bodies primarily because stress granules contain:

(A) Translation initiation factors (eIF4E, eIF3, eIF4G) and small ribosomal subunits
(B) Dcp2 decapping enzyme only
(C) Exosome core components
(D) MicroRNA precursors

96. Selenocysteine, the 21st amino acid, is co-translationally inserted at UGA codons in the presence of:

(A) SECIS hairpin element in mRNA and SelB/eEFSec elongation factor
(B) Amber suppressor tRNA
(C) Puromycin
(D) tmRNA complex

97. Pyrrolysine, the 22nd amino acid found in methanogenic archaea, is decoded at which stop codon?

(A) UAG (Amber)
(B) UAA (Ochre)
(C) UGA (Opal)
(D) AUG

98. Ribosomal frameshifting (-1 programmed frameshifting) is typically triggered by:

(A) A slippery sequence (X_XXY_YYZ) followed by a downstream pseudoknot/hairpin
(B) Absence of release factors
(C) Hyper-osmotic shock
(D) High ATP concentration

99. The pioneer round of eukaryotic translation occurs while the mRNA is bound by:

(A) Nuclear cap-binding heterodimer CBP20/CBP80
(B) Cytoplasmic eIF4E
(C) Poly-A binding protein II
(D) Nucleolin

100. Which transcription factor 'pioneer' is capable of binding tightly closed, compacted heterochromatin to initiate remodeling?

(A) FoxA (HNF3) family
(B) NF-kB
(C) c-Myc
(D) SP1

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the structural organization of transfer RNA (tRNA) and the two-step enzymatic mechanism of tRNA charging by Aminoacyl-tRNA Synthetases.

Q2. Describe Crick's Wobble Hypothesis and explain how non-standard base pairing provides economy and flexibility in codon translation.

Q3. Contrast the structural architecture of the prokaryotic 70S ribosome with the eukaryotic 80S cytosolic ribosome.

Q4. Detail the step-by-step mechanism of translation initiation in prokaryotes, highlighting the functions of IF1, IF2, and IF3.

Q5. Explain the eukaryotic cap-dependent translation initiation pathway, including the 43S Pre-Initiation Complex, scanning, and 80S assembly.

Q6. Describe the translation elongation cycle in bacteria, covering decoding by EF-Tu, peptide bond synthesis, and EF-G-mediated translocation.

Q7. Detail the mechanism of translation termination and ribosome recycling in prokaryotes and eukaryotes.

Q8. Explain the molecular mechanism of action of five major translation-targeting antibiotics (Streptomycin, Tetracycline, Chloramphenicol, Erythromycin, Puromycin).

Q9. Discuss Diphtheria toxin and Ricin as potent biological toxins targeting eukaryotic protein synthesis.

Q10. Explain the molecular regulation of the E. coli lac operon under dual control: negative regulation by Lac repressor and positive regulation by CAP-cAMP.

Q11. Describe the regulation of the tryptophan (trp) operon, contrasting the Trp repressor mechanism with attenuation at the trp leader (trpL).

Q12. Explain the structure and regulatory function of Riboswitches in prokaryotic gene expression.

Q13. Describe the Post-Translational Histone Code hypothesis and explain the epigenetic roles of Histone Acetyltransferases (HATs), Histone Deacetylases (HDACs), and Histone Methyltransferases (HMTs).

Q14. Explain DNA methylation in eukaryotes, detailing the functions of DNMT1, DNMT3A/B, CpG islands, and TET-mediated active demethylation.

Q15. Discuss the architecture of Eukaryotic Promoters, Enhancers, and Insulators, explaining the role of the Mediator complex and Chromatin Looping.

Q16. Compare the structural motifs utilized by transcription factors to recognize DNA: Zinc Finger, Helix-Turn-Helix, Leucine Zipper, and Basic Helix-Loop-Helix.

Q17. Detail the biogenesis, processing, and mechanism of action of MicroRNAs (miRNAs) in eukaryotic post-transcriptional gene regulation.

Q18. Contrast Small Interfering RNAs (siRNAs) with MicroRNAs (miRNAs) in origin, structure, complementarity, and biological silencing mode.

Q19. Explain the functions of Long Non-Coding RNAs (lncRNAs), using the mechanism of X-chromosome inactivation by Xist as a primary paradigm.

Q20. Detail the mechanism of the Integrated Stress Response (ISR) and the molecular switch governing ATF4 translation via upstream open reading frames (uORFs).

Q21. Explain the Ubiquitin-Proteasome System (UPS), covering E1, E2, E3 enzyme cascade, polyubiquitin linkages, and 26S proteasome degradation.

Q22. Describe Nonsense-Mediated mRNA Decay (NMD), detailing how the Exon Junction Complex (EJC) and UPF proteins trigger destruction of aberrant mRNAs.

Q23. Explain the bacterial tmRNA (transfer-messenger RNA) rescue system for ribosomes stalled on non-stop mRNAs.

Q24. Discuss Internal Ribosome Entry Sites (IRES) and explain how viral and cellular mRNAs bypass 5'-cap-dependent translation.

Q25. Explain Co-translational and Post-translational Protein Translocation into the Endoplasmic Reticulum via the Signal Recognition Particle (SRP) and Sec61 translocon.

Q26. Discuss the role of Phase Separation and Membraneless Organelles (Stress Granules and P-bodies) in mRNA compartmentalization and decay.

Q27. Explain the mechanism of Programmed Ribosomal Frameshifting (-1 PRF) and its importance in retroviral and coronavirus gene expression.

Q28. Describe the co-translational insertion of Selenocysteine (the 21st amino acid) and Pyrrolysine (the 22nd amino acid) during genetic decoding.

Q29. Explain Pioneer Transcription Factors and the multi-step mechanism of ATP-Dependent Chromatin Remodeling by the SWI/SNF complex.

Q30. Discuss Epigenetic Inheritance through Mitotic and Meiotic Divisions, highlighting DNA methylation maintenance, histone inheritance, and genomic imprinting.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1: [C]
Q2: [B]
Q3: [A]
Q4: [B]
Q5: [A]
Q6: [A]
Q7: [B]
Q8: [B]
Q9: [B]
Q10: [B]
Q11: [B]
Q12: [A]
Q13: [A]
Q14: [B]
Q15: [A]
Q16: [B]
Q17: [A]
Q18: [B]
Q19: [B]
Q20: [B]
Q21: [A]
Q22: [A]
Q23: [B]
Q24: [A]
Q25: [A]
Q26: [A]
Q27: [B]
Q28: [A]
Q29: [A]
Q30: [A]
Q31: [A]
Q32: [A]
Q33: [A]
Q34: [A]
Q35: [A]
Q36: [A]
Q37: [A]
Q38: [A]
Q39: [A]
Q40: [A]
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Q86: [A]
Q87: [A]
Q88: [A]
Q89: [A]
Q90: [A]
Q91: [A]
Q92: [A]
Q93: [A]
Q94: [A]
Q95: [A]
Q96: [A]
Q97: [A]
Q98: [A]
Q99: [A]
Q100: [A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

[illegible]

[Detailed Step-by-Step Answer]:
Comparative Architecture of 70S and 80S Ribosomes:
1. Prokaryotic 70S Ribosome:
- Molecular Mass: ~2.5 MDa; dimensions ~200 x 220 Angstroms.
- Small Subunit (30S): Composed of 16S rRNA (1,542 nt) and 21 ribosomal proteins (S1-S21). Responsible for decoding and monitoring mRNA-tRNA codon-anticodon fidelity.
- Large Subunit (50S): Composed of 23S rRNA (2,904 nt), 5S rRNA (120 nt), and 31 ribosomal proteins (L1-L36). Contains the catalytic Peptidyl Transferase Center (PTC) and the polypeptide exit tunnel.
2. Eukaryotic 80S Ribosome:
- Molecular Mass: ~4.0 - 4.5 MDa; expanded by eukaryotic-specific ribosomal expansion segments (ES).
- Small Subunit (40S): Composed of 18S rRNA (1,869 nt) and 33 proteins (eS1-eS33). Matches mRNA codons with aminoacyl-tRNA.
- Large Subunit (60S): Composed of 28S rRNA (~4,700 nt), 5.8S rRNA (160 nt), 5S rRNA (120 nt), and 47 proteins (eL1-eL47). Contains the peptidyl transferase ribozyme (28S rRNA) and regulatory surface binding sites for eEFs and signal recognition particles (SRP).

[illegible]

- The TC loads onto the 40S subunit, forming the 43S PIC.

2. mRNA Activation by the eIF4F Complex:

- The eukaryotic mRNA 5' 7-methylguanosine cap is bound by eIF4E.
- eIF4G acts as a master scaffold protein, bridging eIF4E, poly(A)-binding protein (PABP, bound to the 3' poly-A tail), and eIF3, circularizing the mRNA into a 'closed-loop' topology.
- eIF4A (DEAD-box ATP-dependent RNA helicase), assisted by eIF4B, unwinds secondary structures in the 5'-UTR.

3. 48S Complex Formation, 5' -> 3' Scanning, and Start Codon Selection:

- The 43S PIC is recruited to the 5' cap via the eIF3-eIF4G interaction.
- The 43S complex scans linearly along the 5'-UTR in a 5' to 3' direction until it encounters the first AUG embedded within an optimal Kozak consensus sequence (5'-ACCAUGG-3').

4. 80S Assembly:

- Codon recognition triggers eIF5-stimulated GTP hydrolysis by eIF2 and release of Pi.
- eIF5B-GTP promotes joining of the 60S large subunit.
- eIF5B hydrolyzes GTP, resulting in the ejection of eIF1, eIF2-GDP, eIF3, eIF5, and eIF5B, leaving an intact 80S ribosome with Met-tRNA_i^Met in the P-site.

>> [illegible] [illegible] esis, and EF-G-mediated transloc

* RF1 recognizes UAA and UAG.
* RF2 recognizes UAA and UGA.
- Conserved GGQ tripeptide motif of RF1/RF2 enters the peptidyl transferase center and coordinates a water molecule to hydrolyze the ester bond linking the polypeptide to the P-site tRNA, releasing the completed protein.
- Class II RF (RF3-GTP) binds, hydrolyzes GTP, and promotes RF1/RF2 release.
- Ribosome Recycling Factor (RRF) and EF-G-GTP bind the 70S ribosome. GTP hydrolysis by EF-G dissociates 50S and 30S subunits. IF3 binds 30S to eject deacylated tRNA and mRNA.

2. Eukaryotic Termination:
- Omnipotent Class I release factor eRF1 recognizes all three stop codons (UAA, UAG, UGA) through its N-terminal domain.
- eRF1 associates with Class II factor eRF3-GTP. Upon stop codon recognition, eRF3 hydrolyzes GTP, positioning the GGQ motif of eRF1 into the 60S PTC to catalyze ester bond hydrolysis and protein release.
- Ribosome recycling is mediated by the ATP-binding cassette protein ABCE1 (Rli1), which uses ATP hydrolysis to split the 80S ribosome into 60S and 40S subunits.

>> [illegible] cin, Tetracycline, Chloramphenicol,

factor receptor and facilitates endocytic entry and translocation.
- Catalytic Mechanism: Fragment A enters the cytosol and possesses NAD+-diphthamide ADP-ribosyltransferase activity.
- Reaction: Specifically transfers an ADP-ribose moiety from NAD+ onto a post-translationally modified histidine residue (diphthamide) on elongation factor 2 (eEF2):
eEF2 + NAD+ -> ADP-ribosyl-eEF2 + Nicotinamide + H+.
- Result: Inactivated eEF2 can no longer mediate ribosomal translocation. A single molecule of Fragment A can ADP-ribosylate and kill an entire eukaryotic cell.

2. Ricin (Ricinus communis seeds):
- Architecture: Heterodimeric ribosome-inactivating protein (RIP type II). B-chain binds cell surface galactose residues; A-chain is the enzymatic engine.
- Catalytic Mechanism: Ricin A-chain is an RNA N-glycosidase.
- Reaction: In the cytosol, Ricin A cleaves a single specific adenine base (A4324) from a conserved GAGA tetraloop (the sarcin-ricin loop, SRL) of eukaryotic 28S rRNA.
- Result: SRL depurination abolishes the binding and GTPase activation of elongation factors eEF1A and eEF2, completely halting translation.

>>**Solution to Question 14** [illegible]

- Five structural genes (trpE, trpD, trpC, trpB, trpA) encode enzymes for tryptophan biosynthesis.
- Controlled by promoter Ptrp, operator Otrp, and a 162-nt leader region (trpL) containing an attenuator.

2. Trp Repressor-Mediated Negative Control:

- The trpR gene encodes an apo-repressor that cannot bind DNA alone.
- When tryptophan levels are abundant: Two tryptophan molecules bind the apo-repressor as corepressors. This induces a conformational reorientation of its helix-turn-helix motifs, allowing it to bind Otrp and block RNA polymerase binding (~70-fold repression).

3. Attenuation Mechanism (Fine-Tuning Control):

- Operates through tight coupling of transcription and translation within trpL.
- The trpL transcript encodes a 14-amino-acid leader peptide containing two consecutive tryptophan codons (Trp-Trp) and four RNA segments (1, 2, 3, 4) capable of forming mutually exclusive hairpins.
- High Tryptophan: The translating ribosome rapidly traverses the Trp codons and covers segment 2. This leaves segment 3 free to pair with segment 4, forming a stable 3:4 GC-rich hairpin followed by a poly-U tract (intrinsic Rho-independent terminator). RNA polymerase terminates transcription before reaching trpE.
- Low Tryptophan: The ribosome stalls at the tandem Trp codons on segment 1 due to lack of Trp-tRNA^Trp. Segment 2 remains free to base-pair with segment 3, forming a 2:3 antiterminator hairpin. This prevents formation of the 3:4 terminator, allowing RNA polymerase to transcribe the entire structural operon.

[illegible] Histone Acetyltransferases (HATs), [Detailed Step-by-Step Answer]:

The Histone Code Hypothesis and Epigenetic Enzymology:

1. Histone Code Hypothesis:

- Formulated by Strahl and Allis, the hypothesis posits that multiple post-translational covalent modifications (PTMs) on flexible N-terminal histone tails (acetylation, methylation, phosphorylation, ubiquitination) act combinatorially to create a specific molecular barcode.
- This barcode is recognized by effector protein complexes ('readers') that alter higher-order chromatin structure and govern transcriptional access.

2. Histone Acetylation Dynamics:

- Histone Acetyltransferases (HATs, e.g., p300/CBP, Gcn5): Transfer acetyl groups from acetyl-CoA to the epsilon-amino group of lysine residues (e.g., H3K9ac, H3K27ac). Neutralizing the positive charge abolishes electrostatic interactions with negatively charged DNA, decondensing chromatin into open euchromatin. Acetyl-lysines are recognized by Bromodomain-containing readers.
- Histone Deacetylases (HDACs, Classes I-IV): Catalyze the hydrolytic removal of acetyl groups, restoring lysine positive charge, promoting chromatin compaction into heterochromatin, and repressing transcription.

3. Histone Methylation Dynamics:

- Histone Methyltransferases (HMTs): Transfer 1, 2, or 3 methyl groups from S-adenosylmethionine (SAM) to lysine or arginine residues without altering overall charge.
- Functional Dualism:

* Activating Marks: H3K4 trimethylation (H3K4me3) at promoters and H3K36me3 in actively transcribed gene bodies.
* Repressive Marks: H3K9me3 (constitutive heterochromatin, bound by HP1 chromodomain) and H3K27me3 (facultative heterochromatin, catalyzed by PRC2/EZH2 and bound by PRC1 chromodomain).

>> Solution to Question 14:

[Problem Statement]: Explain DNA methylation in eukaryotes, detailing the functions of DNMT1, DNMT3A/B, CpG islands, and TET-mediated active demethy

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[illegible]

variable length. Forms homo- or heterodimers that bind E-box consensus sequences (5'-CANNTG-3') via basic N-terminal segments.
- Examples: MyoD (muscle determination), c-Myc, Max.

[illegible]

- miRNA: Primarily induces translational repression, deadenylation, decapping, and exonucleolytic decay.
- siRNA: AGO2 catalytically slices the target mRNA backbone at the phosphodiester bond opposite nucleotides 10 and 11 of the siRNA guide strand ('slicer activity'), causing immediate and complete destruction of the target transcript.

[illegible]

3. Paradoxical Activation of ATF4 Translation:

- While global translation drops, activating transcription factor 4 (ATF4) translation is dramatically induced.
- The ATF4 5'-UTR contains two upstream open reading frames: uORF1 (positive regulatory element) and uORF2 (inhibitory element overlapping the ATF4 coding frame out-of-phase).
- Under Non-Stressed Conditions (Abundant Ternary Complex): Ribosomes translate uORF1, rapidly re-acquire a new ternary complex, initiate at uORF2, terminate past the ATF4 start codon, and dissociate. Zero ATF4 is translated.
- Under Stressed Conditions (Scarce Ternary Complex): Ribosomes translate uORF1 and resume scanning, but because ternary complex is scarce, re-initiation is delayed. The scanning 40S subunit bypasses the inhibitory uORF2 start codon and finally acquires the ternary complex just in time to initiate at the real ATF4 coding sequence.
- ATF4 activates adaptive genes for amino acid transport, redox balance, and autophagy.

[illegible]

1. Biological Purpose:
- An essential eukaryotic mRNA surveillance pathway that detects and degrades transcripts harboring premature termination codons (PTCs), preventing the accumulation of truncated, potentially dominant-negative proteins.

2. The Exon Junction Complex (EJC) Flag:
- During pre-mRNA splicing in the nucleus, a multi-protein complex called the EJC (composed of eIF4A3, MAGOH, Y14, and MLN51) is deposited exactly 20-24 nucleotides upstream of every exon-exon junction.
- EJCs remain bound to mRNA during nuclear export into the cytoplasm.

3. The Pioneer Round of Translation and Rule of 50-55 Nucleotides:
- During the initial 'pioneer round' of translation (mRNA bound to CBP80/CBP20), the moving 80S ribosome physically strips EJCs off the transcript.
- Normal Stop Codon: Located in the terminal exon; therefore, no EJCs remain downstream when the ribosome terminates.
- Premature Termination Codon (PTC): If a stop codon occurs more than 50-55 nucleotides upstream of the final exon junction, the ribosome terminates prematurely while one or more EJCs remain downstream.

4. NMD Activation and mRNA Destruction:
- Stalled ribosome at the PTC recruits release factors eRF1/eRF3, which recruit the master NMD helicase UPF1.
- Downstream EJC-associated factors UPF2 and UPF3 bridge to UPF1, forming the SURF complex (SMG1-UPF1-eRF1-eRF3).
- SMG1 kinase phosphorylates UPF1.
- Phospho-UPF1 triggers translation repression and recruits SMG6 (endonuclease that cleaves mRNA at the PTC) and SMG5/SMG7, which recruit decapping (DCP2) and deadenylation complexes for rapid 5'->3' (XRN1) and 3'->5' (Exosome) degradation.

>> Solution to Question 23:

[Problem Statement]: Explain the bacterial tmRNA (transfer-messenger RNA) rescue system for ribosomes stalled on non-stop mRNAs.

- The 70S ribosome is disassembled and recycled by RRF and EF-G.
- The damaged non-stop mRNA is degraded by RNase R.
- The C-terminal tag (-A-A-COOH) is recognized by ClpXP and ClpAP ATP-dependent proteases, ensuring immediate destruction of the truncated defective protein.

[illegible]

coordinated GTP hydrolysis triggers release of SRP, transferring the ribosome-nascent chain to the Sec61 Translocon.
4. Translocation and Processing:
- Sec61 is a heterotrimeric membrane channel (alpha, beta, gamma) featuring a central hourglass pore and a lateral gate.
- Resumed translation feeds the growing chain through the hydrophilic channel into the ER lumen.
- Signal Peptidase cleaves the signal peptide.
- Oligosaccharyltransferase (OST) transfers a pre-assembled Glc3Man9GlcNAc2 oligosaccharide onto consensus Asn-X-Ser/Thr motifs (N-linked glycosylation), and ER chaperones (BiP/GRP78, Calnexin) guide proper folding.

[illegible]

3. Biophysical Slipping Mechanism:
- As the ribosome translocates along the slippery sequence, its entry into the mRNA tunnel is physically blocked by the unwinding resistance of the stable downstream pseudoknot.
- This mechanical tension pauses the ribosome over the slippery site. During the pause, the tRNA anticodons in the A and P sites detach from their original codons (XXY and YYZ) and slip backward by 1 base to pair with the -1 frame codons (XXX and YYY), maintaining at least 2 out of 3 base pairs.
- Once translocated in the -1 frame, the ribosome resumes translation across the unfolded pseudoknot.

4. Viral Examples and Stoichiometric Balance:
- Retroviruses (e.g., HIV-1): Uses -1 PRF to express the Gag-Pol polyprotein at a fixed ~5-10% ratio relative to the structural Gag protein (~20:1 ratio required for viral capsid and enzyme assembly).
- Coronaviruses (e.g., SARS-CoV-2): Utilizes -1 PRF at the boundary of ORF1a and ORF1b to express the RNA-dependent RNA polymerase (RdRp / nsp12) and other viral replication machinery.

[illegible]

- Standard transcription factors cannot access their recognition sequences when DNA is wrapped around histone octamers in condensed heterochromatin.
- Pioneer Factors (e.g., FoxA/HNF3, GATA, Oct4, Sox2): Possess specialized winged-helix or HMG structural domains that recognize target sequence motifs exposed on the outer surface of nucleosomal DNA.
- They bind tightly closed heterochromatin without displacing histones initially, serving as a beacon that recruits ATP-dependent chromatin remodelers and histone acetyltransferases.

2. ATP-Dependent Chromatin Remodeling:

- Large multi-protein machines that use energy from ATP hydrolysis to alter histone-DNA contacts, classified into four major families: SWI/SNF, ISWI, CHD, and INO80.

3. Mechanism of the SWI/SNF (BAF/PBAF) Complex:

- Core Catalytic Engine: Contains an essential SWI2/SNF2-family DNA translocase ATPase subunit (SMARCA4/BRG1 or SMARCA2/BRM) and bromodomains.
- DNA Translocation and Loop Propagation:

* The translocase anchors onto the nucleosomal core particle at superhelical location 2 (SHL2).
* Consuming 1 ATP per base pair, it pumps DNA around the histone octamer, generating local torsional strain and peeling a wave of DNA off the histone surface (DNA loop propagation).
* This pushes or 'slides' the histone octamer along the DNA fiber or completely evicts histone H2A-H2B dimers or the full octamer.

- Result: Exposes previously occluded promoters and enhancers, allowing basal transcription factors (TFIID) and RNA Polymerase II to dock and initiate transcription.

>> Solution to Question 30:

[Problem Statement]: Discuss Epigenetic Inheritance through Mitotic and Meiotic Divisions, highlighting DNA methylation maintenance, histone inheritance, a